Supplementary Information: Molecular Skala in CP2K

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1 Molecular Density Representations

The molecular interface covers three density representations. GPW-GTH supplies the AO valence density of the GTH pseudopotential Hamiltonian. GAPW-AE, selected with `POTENTIAL ALL`, supplies the full AO density. GAPW-GTH and GAPW-ECP supply the AO valence density of the corresponding effective Hamiltonian. In every case, CP2K passes the geometry, Gaussian basis, spin-resolved AO density matrix, and communicator to GauXC, which returns the XC energy, AO XC matrix, and available nuclear derivatives.

The direct molecular GauXC route does not include a nonlinear core correction (NLCC), because the frozen-core density and its derivatives are not part of the AO density-matrix interface. Higher XC response kernels are likewise outside the present scope.

2 Small-Molecule Computational Details

All small-molecule validation calculations use isolated molecular boundary conditions in Quickstep. They follow the MOLOPT_UZH protocol[1] with $E_{\mathrm{cut}} = 600$ Ry and $E_{\mathrm{rel}} = 60$ Ry. GPW-GTH uses TZV2P molecularly optimized basis sets and matching analytic PBE GTH pseudopotentials. GAPW-AE uses QZVPP all-electron MOLOPT_UZH basis sets, `POTENTIAL ALL`, and `GAPW_ACCURATE_XCINT` for the native PBE reference. GAPW-GTH and GAPW-ECP use the AO valence density of their corresponding effective Hamiltonians; HCl provides the GAPW-ECP diagnostic.

Within each native-PBE/PBE-through-GauXC pair, the geometry, basis, effective Hamiltonian, spin convention, SCF threshold, and numerical grid are identical. Unless stated otherwise, energy differences are reported in hartree, force errors in hartree/bohr, and molecular-virial errors in hartree.

3 AO Matrices and Parallel Execution

CP2K stores AO matrices in DBCSR form. For a GauXC evaluation, each MPI rank expands its local blocks, an all-reduction constructs the replicated dense density matrix, and GauXC returns a dense XC matrix. The latter is symmetrized before insertion into a DBCSR matrix with the full upper-triangular block structure, since a quadrature-generated XC matrix need not share the thresholded sparsity pattern of the input density.

For unrestricted calculations the spin-channel matrices are transformed as

$$P^s = P^\alpha + P^\beta, \qquad P^z = P^\alpha - P^\beta, \tag{S1}$$

and the returned derivatives are transformed back with $V^\alpha = V^s + V^z$ and $V^\beta = V^s - V^z$.

The GauXC runtime is created on the communicator that owns the current force evaluation. This distinction matters for CDFT and CDFT-CI calculations, where diabatic states can occupy disjoint MPI subgroups. Communicator-local grid and integrator objects prevent collectives for one subgroup from using counts or displacements belonging to another. The current implementation caches compatible GauXC objects across repeated SCF evaluations and rebuilds them only when the basis, geometry, quadrature, execution space, or communicator changes.

4 Finite-Difference Checks

Forces are validated against central finite differences of the total energy,

$$F_{A,i}^{\mathrm{FD}} = -\frac{E(R_{A,i}+\delta) - E(R_{A,i}-\delta)}{2\delta}. \tag{S2}$$

The displacement is chosen small enough to probe the local derivative and large enough to remain above SCF and quadrature noise.

For the molecular virial, coordinates are deformed affinely around a fixed center $\mathbf{R}_0$,

$$R_{A,k}^{(\pm;ij)} = R_{0,k} + (R_{A,k} - R_{0,k}) \pm \delta\, \delta_{ki}(R_{A,j} - R_{0,j}), \tag{S3}$$

and each tensor component is checked as

$$\Xi_{ij}^{\mathrm{FD}} = \frac{E^{(+;ij)} - E^{(-;ij)}}{2\delta}. \tag{S4}$$

The scalar values tabulated below use the corresponding isotropic coordinate scaling. These are molecular coordinate-scaling diagnostics and must not be interpreted as periodic cell stresses.

5 Complete Small-Molecule Validation

Table S1 collects the complete molecular diagnostics used to construct the compact main-manuscript table. It gives the PBE interface comparison and the PBE and Skala finite-difference errors for the H_2, NH_3, H_2O, and HCl calculations. Individual GAPW-GTH and GAPW-ECP total energies are not compared across effective Hamiltonians because their energy zero depends on the chosen pseudopotential or ECP; the transferable validation quantities are matched native/external-library energy differences and derivatives.

Table S1: Complete small-molecule validation. $\Delta E_{\mathrm{PBE}} = E_{\mathrm{PBE}}^{\mathrm{GauXC}} - E_{\mathrm{PBE}}^{\mathrm{CP2K}}$ is in hartree. Force errors are in hartree/bohr and molecular-virial errors in hartree.

System	Method	ΔE_{PBE}	$\lvert\Delta F_{\mathrm{PBE}}\rvert$	$\lvert\Delta W_{\mathrm{PBE}}\rvert$	$\lvert\Delta F_{\mathrm{Skala-1.1}}\rvert$	$\lvert\Delta W_{\mathrm{Skala-1.1}}\rvert$
H_2	GPW-GTH	1.27×10^{-7}	7.60×10^{-7}	3.41×10^{-9}	2.60×10^{-7}	3.01×10^{-6}
NH_3	GPW-GTH	1.08×10^{-5}	7.60×10^{-8}	1.97×10^{-7}	9.86×10^{-8}	5.07×10^{-6}
H_2O	GPW-GTH	8.93×10^{-5}	4.69×10^{-7}	4.69×10^{-7}	9.17×10^{-7}	1.09×10^{-5}
H_2	GAPW-GTH	1.74×10^{-7}	7.60×10^{-7}	3.41×10^{-9}	6.44×10^{-7}	3.21×10^{-6}
NH_3	GAPW-GTH	5.76×10^{-6}	1.23×10^{-7}	2.03×10^{-7}	5.27×10^{-8}	6.45×10^{-6}
H_2O	GAPW-GTH	1.13×10^{-4}	4.69×10^{-7}	1.43×10^{-7}	2.41×10^{-6}	1.24×10^{-5}
H_2	GAPW-AE	1.11×10^{-6}	7.65×10^{-7}	3.56×10^{-9}	9.47×10^{-7}	1.63×10^{-6}
NH_3	GAPW-AE	-1.47×10^{-5}	7.52×10^{-8}	1.73×10^{-7}	8.23×10^{-6}	3.40×10^{-5}
H_2O	GAPW-AE	-2.94×10^{-5}	4.70×10^{-7}	3.78×10^{-7}	1.99×10^{-5}	7.25×10^{-6}
HCl	GAPW-ECP	2.26×10^{-5}	5.38×10^{-7}	8.13×10^{-9}	9.96×10^{-6}	3.00×10^{-6}

6 Meta-GGA Kinetic-Energy-Density Validation

The Skala descriptor path depends explicitly on the KS kinetic-energy density τ. We therefore validate this ingredient independently with closed-shell H_2O using TPSS[2] and r^{2}SCAN.[3] Table S2 compares the standard 600/60 Ry grid with a tighter 1200/80 Ry grid. For GPW-GTH, tightening the grid reduces the native-CP2K force and molecular-virial finite-difference errors by approximately two orders of magnitude, whereas the corresponding GauXC derivative errors and XC-only virial residuals are already stable. The GAPW-AE derivatives show substantially weaker grid sensitivity. The comparison therefore identifies the dominant effect as the slower native-grid convergence of the GPW representation for τ-dependent functionals; it does not imply monotonic convergence of every GAPW-AE energy difference.

Table S2: Closed-shell H_2O meta-GGA kinetic-energy-density convergence diagnostic. $\Delta E = E^{\text{GauXC}} - E^{\text{CP2K}}$ is in hartree; force errors are in hartree/bohr and molecular-virial errors in hartree.

Method	Functional	ΔE	$\lvert\Delta F\rvert_{\text{CP2K}}$	$\lvert\Delta F\rvert_{\text{GauXC}}$	$\lvert\Delta W\rvert_{\text{CP2K}}$	$\lvert\Delta W\rvert_{\text{GauXC}}$	$\lvert\Delta W_{\text{xc}}\rvert_{\text{GauXC}}$
(a) Standard grid: $E_{\text{cut}} = 600$ Ry and $E_{\text{rel}} = 60$ Ry							
GPW-GTH	TPSS	-7.019×10^{-4}	1.29×10^{-5}	4.84×10^{-7}	7.38×10^{-5}	4.44×10^{-7}	1.05×10^{-8}
GPW-GTH	r^{2}SCAN	-1.805×10^{-4}	1.23×10^{-5}	4.88×10^{-7}	5.46×10^{-5}	1.78×10^{-7}	9.92×10^{-9}
GAPW-AE	TPSS	-5.222×10^{-4}	5.14×10^{-7}	5.19×10^{-7}	3.35×10^{-7}	3.21×10^{-7}	1.15×10^{-8}
GAPW-AE	r^{2}SCAN	$+9.28\times10^{-5}$	5.30×10^{-7}	5.22×10^{-7}	4.13×10^{-8}	3.04×10^{-8}	1.18×10^{-8}
(b) Tight grid: $E_{\text{cut}} = 1200$ Ry and $E_{\text{rel}} = 80$ Ry							
GPW-GTH	TPSS	-3.309×10^{-4}	4.76×10^{-7}	4.85×10^{-7}	4.42×10^{-7}	4.46×10^{-7}	1.06×10^{-8}
GPW-GTH	r^{2}SCAN	-8.230×10^{-5}	5.02×10^{-7}	4.89×10^{-7}	9.52×10^{-8}	1.80×10^{-7}	9.82×10^{-9}
GAPW-AE	TPSS	-4.064×10^{-4}	5.14×10^{-7}	5.18×10^{-7}	3.40×10^{-7}	3.22×10^{-7}	1.18×10^{-8}
GAPW-AE	r^{2}SCAN	$+1.164\times10^{-4}$	5.17×10^{-7}	5.22×10^{-7}	5.55×10^{-8}	3.15×10^{-8}	1.19×10^{-8}

7 Water-Hexamer Benchmark Details

The water-hexamer calculations provide a complementary many-body stress test rather than a second broad benchmark. The low-lying prism, cage, book, bag, cyclic-chair, and cyclic-boat isomers probe small relative-energy splittings controlled by cooperative polarization, many-body exchange repulsion, charge redistribution, and dispersion-like nonlocal correlation.[4, 5, 6, 7] The structures and CCSD(T)/CBS no-counterpoise reference binding energies are taken from the updated water-cluster subset of the Benchmark Energy and Geometry Database (BEGDB) and are used without zero-point corrections.[8, 7]

The primary protocol uses GAPW-AE, selected with `POTENTIAL ALL`, QZVPP-quality MOLOPT_UZH basis sets,[1] and `GAPW_ACCURATE_XCINT`. The latter option retains the hard all-electron and soft compensation one-center contributions in the native PBE XC integration and therefore provides the appropriate native CP2K comparison for the all-electron AO density evaluated by GauXC. PBE-through-GauXC and native PBE use the same structures, basis, spin convention, SCF thresholds, and numerical grids.

The PBE-D3(BJ) values are native CP2K calculations with the Becke–Johnson damped D3 correction.[9, 10] Following the public Skala-1.1 model metadata, the Skala-1.1-D3(BJ) values add the B3LYP5-D3(BJ) binding contribution as a separate post-SCF correction because the present interface keeps `XC_FUNCTIONAL/GAUXC` and `VDW_POTENTIAL` separate.

For isomer i, the binding energy, relative energy, and Skala relative-energy error are

$$E_{b,i} = E_i^{(\mathrm{H_2O})_6} - 6E^{\mathrm{H_2O}}, \tag{S5}$$

$$\Delta E_i^{\text{hex}} = E_{b,i} - \min_j E_{b,j}, \tag{S6}$$

$$\delta_i^{\text{hex}} = \Delta E_{i,\text{Skala}}^{\text{hex}} - \Delta E_{i,\text{ref}}^{\text{hex}}. \tag{S7}$$

Referencing each method to its own lowest-energy isomer removes absolute-energy offsets and focuses the comparison on the ordering within a fixed Hamiltonian and density representation.

The GauXC-PBE and native PBE columns agree at the level expected from the residual difference between the atom-centered GauXC quadrature and the native augmented XC integration. Skala-1.1 reduces the mean unsigned relative-energy error from 6.10 to 5.44 kcal mol^{-1}, but it still overstabilizes the compact cage and prism family relative to the book and cyclic structures. Adding the B3LYP5-D3(BJ) contribution improves the prism–cage splitting but increases the mean unsigned error over all eight isomers to 6.79 kcal mol^{-1}. These results are therefore interpreted as a demanding many-body stress test rather than as evidence of uniform improvement for hydrogen-bonded networks.

Table S4 summarizes the protocol sensitivity. The QZVPP GAPW-AE row corresponds to the primary data in Table S3; the TZVPP GAPW-AE protocol probes basis-set sensitivity, and the GPW-GTH protocol probes the same workflow in a pseudopotential representation. The maximum PBE mismatch is the largest absolute difference between GauXC-PBE and native CP2K PBE relative energies over the eight isomers.

Table S5 gives the complementary GPW-GTH calculation using PBE-optimized molecular basis sets and

Table S3: Primary GAPW-AE water-hexamer relative binding energies without zero-point corrections. All values are in kcal mol^{-1} relative to the lowest isomer within the respective method. The reference column reports the BEGDB CCSD(T)/CBS no-counterpoise values. The calculations use `POTENTIAL ALL`, QZVPP-quality MOLOPT_UZH basis sets, and `GAPW_ACCURATE_XCINT`.

Isomer	Ref.	GauXC-PBE	PBE	PBE-D3(BJ)	Skala-1.1	Skala-1.1-D3(BJ)
Prism	0.00	0.55	0.54	0.33	0.37	0.06
Cage	0.21	0.00	0.00	0.00	0.00	0.00
Book 1	0.63	10.26	10.25	11.10	9.82	11.13
Book 2	1.00	5.17	5.16	5.93	4.86	6.06
Cyclic chair	1.54	16.21	16.21	17.90	14.92	17.52
Bag	1.55	1.90	1.90	2.48	1.70	2.66
Cyclic boat 1	2.57	9.20	9.20	10.83	7.82	10.34
Cyclic boat 2	2.63	15.28	15.28	16.90	13.71	16.22
MUE	–	6.11	6.10	6.97	5.44	6.79

Table S4: Water-hexamer protocol sensitivity. All entries except the maximum PBE mismatch are mean unsigned relative-energy errors in kcal mol^{-1} with respect to the BEGDB reference.

Protocol	$\max\lvert\Delta E_{\mathrm{PBE}}\rvert$	GauXC-PBE	PBE	PBE-D3(BJ)	Skala-1.1	Skala-1.1-D3(BJ)
GPW-GTH PBE	0.125	6.11	6.08	6.94	5.62	6.97
GAPW-AE TZVPP	0.008	5.83	5.83	6.66	5.17	6.41
GAPW-AE QZVPP	0.009	6.11	6.10	6.97	5.44	6.79

matching PBE GTH pseudopotentials. It documents the numerical behavior of the same CP2K–GauXC interface with an effective valence Hamiltonian.

Table S5: Complementary GPW-GTH water-hexamer relative binding energies with PBE-optimized molecular basis sets and PBE GTH pseudopotentials. All values are in kcal mol^{-1}, relative to the lowest isomer within the respective method.

Isomer	Ref.	GauXC-PBE	PBE	PBE-D3(BJ)	Skala-1.1	Skala-1.1-D3(BJ)
Prism	0.00	1.18	1.11	0.90	0.92	0.61
Cage	0.21	0.00	0.00	0.00	0.00	0.00
Book 1	0.63	10.30	10.29	11.14	10.10	11.41
Book 2	1.00	5.45	5.45	6.21	5.24	6.45
Cyclic chair	1.54	15.55	15.42	17.11	14.60	17.20
Bag	1.55	2.21	2.20	2.78	2.10	3.06
Cyclic boat 1	2.57	8.91	8.84	10.47	7.92	10.45
Cyclic boat 2	2.63	14.97	15.03	16.65	13.77	16.28
MUE	–	6.11	6.08	6.94	5.62	6.97

Table S6 gives the GAPW-AE TZVPP basis-set check. It reproduces the same qualitative pattern as the QZVPP protocol: close GauXC-PBE/native-PBE agreement, a moderate reduction of the mean unsigned error from Skala-1.1, and no uniform improvement from the additive D3(BJ) correction.

8 Computational Details for the dietGMTKN55 Validation

Single-point energies were evaluated at the fixed dietGMTKN55 geometries. The same Skala-1.1 model was used in the reference PySCF implementation and, after conversion to the GauXC model format, in CP2K. The calculations used spherical def2-TZVP basis functions [11]; the minimally augmented ma-def2-TZVP basis [12] was selected for the dietGMTKN55 subsets that included anions. All Skala calculations included

Table S6: Auxiliary GAPW-AE TZVPP water-hexamer relative binding energies for basis-set sensitivity. All values are in kcal mol^{-1}, relative to the lowest isomer within the respective method.

Isomer	Ref.	GauXC-PBE	PBE	PBE-D3(BJ)	Skala-1.1	Skala-1.1-D3(BJ)
Prism	0.00	0.78	0.77	0.56	0.64	0.33
Cage	0.21	0.00	0.00	0.00	0.00	0.00
Book 1	0.63	9.88	9.88	10.72	9.33	10.64
Book 2	1.00	4.72	4.72	5.48	4.29	5.50
Cyclic chair	1.54	15.64	15.64	17.33	14.23	16.83
Bag	1.55	1.41	1.40	1.99	1.09	2.05
Cyclic boat 1	2.57	8.76	8.76	10.39	7.35	9.87
Cyclic boat 2	2.63	14.86	14.86	16.48	13.25	15.76
MUE	–	5.83	5.83	6.66	5.17	6.41

the Becke–Johnson damped D3 correction with B3LYP5 settings [9, 10].

The CP2K calculations used CP2K 2026.2 [13], with GAPW-AE for the light-element systems and GAPW-ECP for the heavy-element cases specified below. The GauXC quadrature used the `FINE` grid, `ROBUST` pruning, and the Mura–Knowles radial quadrature. The plane-wave cutoff and relative cutoff were 500 and 50 Ry, respectively, with four multigrid levels. `EPS_SCF` and `EPS_DEFAULT` were 10^{-5} and 10^{-10}, respectively, with a maximum of 100 SCF iterations. The SCF equations were solved by diagonalization using direct density-matrix mixing with a mixing factor of 0.4 and two previous densities. Each molecule was centered in a non-periodic orthorhombic cell with at least 15 Å between the nearest atom and every cell face. All calculations used the analytic Poisson solver. Four reactions, comprising eight unique molecules containing Bi, Te, or I, used the corresponding def2 effective-core potentials [14, 15].

The reference calculations used PySCF 2.10.0 [16] with its level-3 Treutler–Ahlrichs atom-centered grid, original Becke partitioning without an atomic-radius adjustment, NWChem pruning, and a small-density cutoff of 10^{-8}. The SCF energy and gradient thresholds were 5×10^{-6} E_h and 10^{-3}, respectively, with a maximum of 60 iterations.

Figure 1 gives the direct reaction-by-reaction cross-code verification. The mean absolute difference between the CP2K/GauXC and PySCF reaction energies is 0.108 kcal mol^{-1}, and the mean signed difference $E_{\mathrm{CP2K}} - E_{\mathrm{PySCF}}$ is -0.040 kcal mol^{-1}. With the exception of the RC21 diethyl-ether radical-cation cleavage (A) and the BH76 chloride identity-$\mathrm{S_N}2$ barrier (B), all absolute cross-code differences are smaller than 0.7 kcal mol^{-1}. The two exceptions are 2.739 and 0.969 kcal mol^{-1}, respectively.

9 CPU/GPU Timing Diagnostic

Table S7 reports a controlled molecular GauXC/Skala hardware-path diagnostic using CP2K source revision `21ef8686db`, the optimized Skala-1.1 Rev1 host and CUDA model artifacts, and an NVIDIA GB10 system. The isolated GPW-GTH water clusters were evaluated with one MPI rank and ten OpenMP threads, one SCF Kohn–Sham matrix build from an atomic guess, $E_{\mathrm{cut}} = 150$ Ry, $E_{\mathrm{rel}} = 30$ Ry, and the GauXC `FINE/ROBUST` quadrature. The deliberately lightweight numerical settings isolate the host/device execution paths and are not the production-accuracy protocol used for the molecular validation. One warm-up run and three measured runs were performed for each size and backend.

The accelerator benefit grows with system size: the GPU is slower for the monomer because fixed initialization costs dominate, but reaches wall-time and Kohn–Sham-matrix speedups of 3.28 and 3.83, respectively, for the octamer. Relative to the earlier measurement with the same diagnostic protocol, current CPU wall times are 15–25% lower. GPU wall time is essentially unchanged for the octamer, while the smaller systems expose a larger fixed startup contribution. A same-build comparison with the original model artifacts changes GPU wall times by no more than approximately 3%, with a 2–3% reduction for the larger clusters; the Rev1 artifacts therefore do not cause the reduced small-system speedup. The median peak host resident-set sizes for CPU/GPU execution are 1.94/2.84, 3.34/2.96, 5.77/3.06, and 9.95/3.09 GiB for $n = 1, 2, 4, 8$, respectively. The unified-memory GB10 platform does not expose a separate device-memory

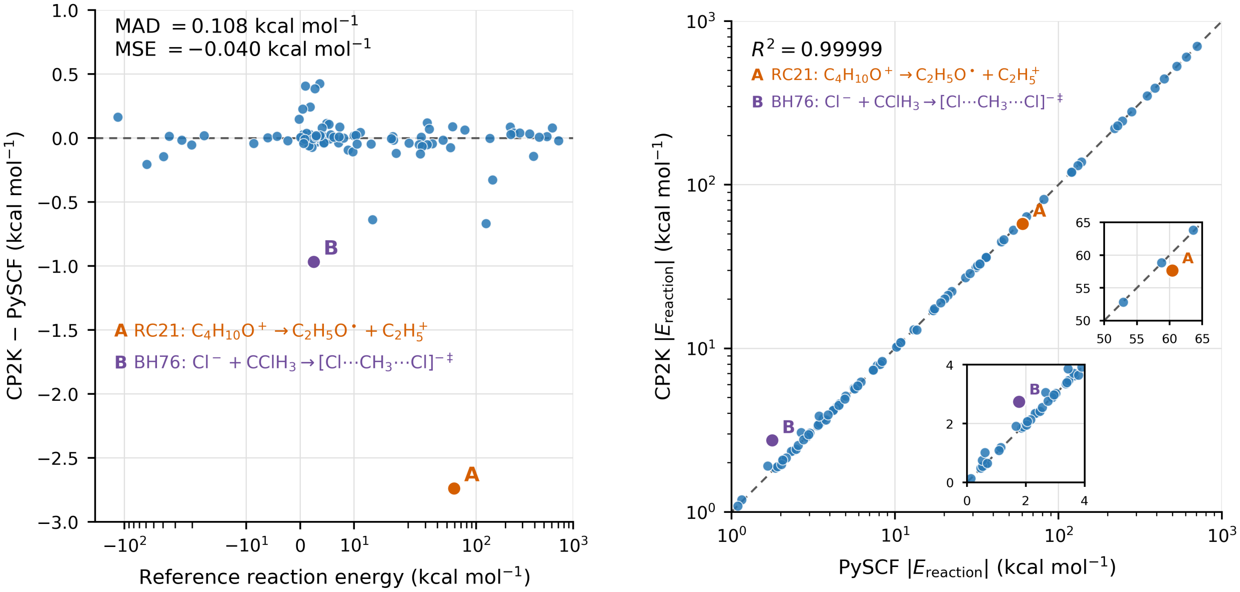

Figure 1: Reaction-by-reaction verification of the dietGMTKN55 energies. Left: signed CP2K/GauXC–PySCF reaction-energy differences plotted against the reference reaction energies, using a symmetric logarithmic horizontal axis. Right: absolute reaction energies from the two implementations on logarithmic axes. A denotes the RC21 diethyl-ether radical-cation cleavage and B the BH76 chloride identity-S_N2 barrier.

Table S7: Current Spark CPU/GPU Skala-through-GauXC timing diagnostic for isolated $(H_2O)_n$ clusters. Times are in seconds and are medians over three measured runs after one warm-up run. The energy difference is $E_{\mathrm{GPU}} - E_{\mathrm{CPU}}$.

n	Atoms	$t_{\mathrm{wall}}^{\mathrm{CPU}}$	$t_{\mathrm{wall}}^{\mathrm{GPU}}$	S_{wall}	$t_{\mathrm{KS}}^{\mathrm{CPU}}$	$t_{\mathrm{KS}}^{\mathrm{GPU}}$	S_{KS}	ΔE (Ha)
1	3	3.37	3.72	0.91	1.89	1.99	0.95	1.24×10^{-8}
2	6	5.54	4.71	1.18	3.95	2.72	1.45	2.94×10^{-8}
4	12	13.42	5.73	2.34	11.62	3.58	3.24	7.55×10^{-8}
8	24	41.54	12.66	3.28	39.69	10.37	3.83	1.48×10^{-7}

counter through `nvidia-smi`; consequently, only host resident-set size is reported. These single-rank, single-build measurements quantify the current hardware paths but do not constitute an MPI-scaling benchmark.

10 Data Package

The companion repository `DCM-Uni-Paderborn/Molecular-Skala-in-CP2K` contains the molecular structures, CP2K inputs, selected raw outputs, processed tables, checksums, and short extraction scripts needed to reproduce the numerical tables in the manuscript. The current Spark CPU/GPU timing diagnostic reported above is included there as raw run summaries, source and model metadata, and the processed comparison table `processed/gpu_timing_current_comparison_20260813.tsv`; the earlier timing summaries are retained for provenance. Diagnostic development files that are not required for reproducing the published tables are kept out of the Supplementary Information and, where retained for provenance, are clearly separated in the repository.

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